Shopify Stock Data – Data Modeling, DAX and Time Intelligence

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Programme: B.Sc. Data Science and AI
Tool Used: Microsoft Power BI Desktop
Dataset: Shopify Stock Dataset

1. Introduction:

Week 3 focused on preparing and modelling Shopify stock data using Microsoft Power BI.

The work included importing and cleaning the stock dataset, creating a dedicated date table using DAX, establishing a relationship between the date table and stock table, creating a custom date hierarchy, and developing financial and time-intelligence measures using DAX.

The main objective was to create a structured data model that could support financial and time-based analysis of Shopify stock prices and trading volume.

2. Importing and Preparing the Shopify Stock Data

The Shopify stock dataset was imported into Power BI and opened in Power Query Editor for data preparation.

The dataset contained the following seven columns:

- date
- open
- high
- low
- close

- adj_close
- volume

The column data types were checked during the preparation process. The date column was converted from its original Date/Time/Timezone format to **Date** using the **Date Only** transformation. The numerical stock price columns were maintained as numeric values, while volume was maintained as a whole number.

The data quality indicators showed **100% Valid, 0% Error and 0% Empty** for the displayed columns. After completing the preparation, the data was loaded into Power BI using **Close & Apply**.

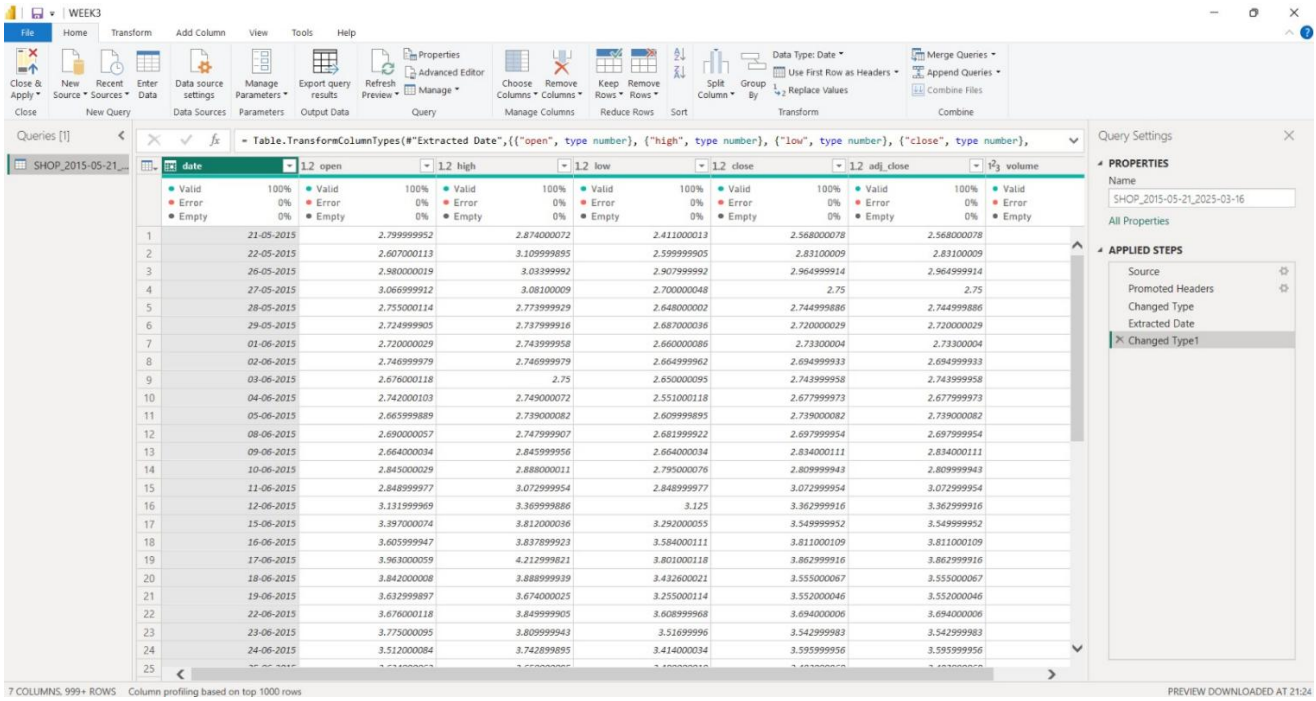


Figure 1: Shopify Stock dataset after data preparation in Power Query

3. Creating the Dim_Date Table:

A separate date dimension was created to support time-based analysis.

The Dim_Date table was generated using the DAX CALENDARAUTO() function.

Dim_Date = CALENDARAUTO()

This automatically generated a continuous date table based on the dates available in the Power BI model.

The Dim_Date table was then marked as the official **Date Table** in Power BI using the Date column. This allows the date table to be used with Power BI's time-intelligence calculations.

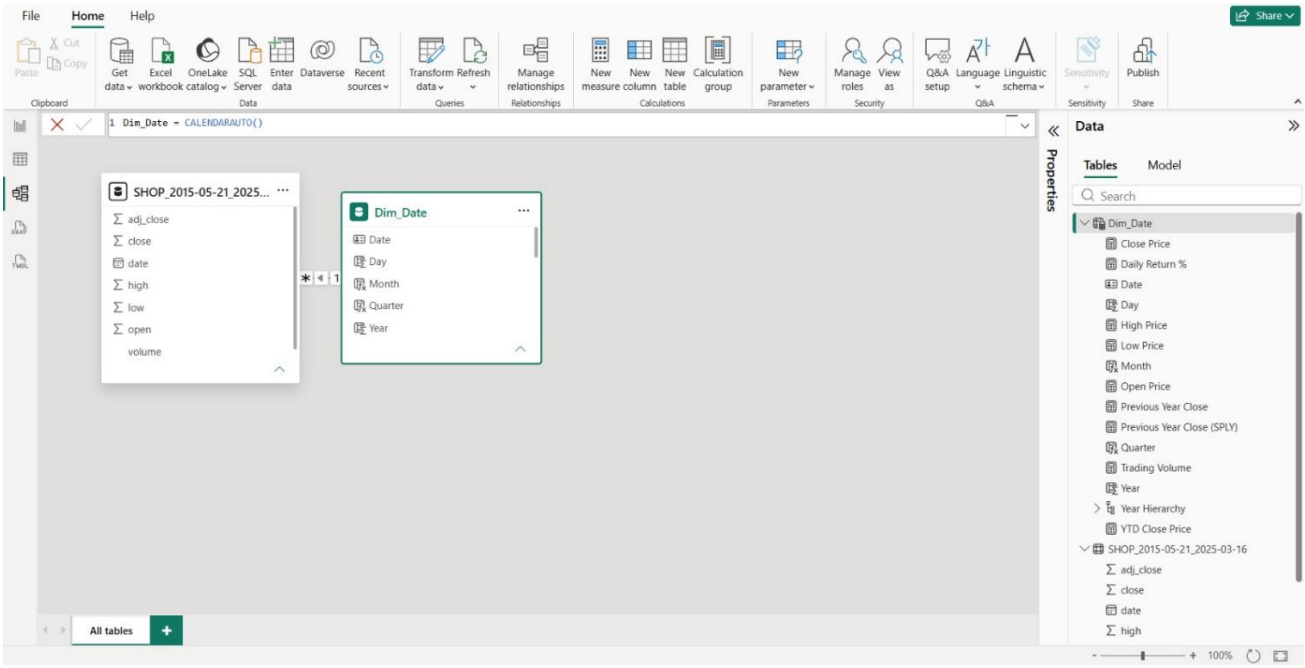


Figure 2: Creation of the Dim_Date table using CALENDARAUTO()

4. Creating Date Attributes and Custom Hierarchy:

Additional date attributes were created in the Dim_Date table to support time-based analysis.

The following columns were created:

Year

Year = YEAR('Dim_Date'[Date])

Quarter

Quarter = "Q" & QUARTER('Dim_Date'[Date])

Month

Month = FORMAT('Dim_Date'[Date], "MMMM")

Day

Day = DAY('Dim_Date'[Date])

A custom date hierarchy named **Year Hierarchy** was then created. The hierarchy was arranged in the following order:

Year → Quarter → Month → Day

This hierarchy can be used for drill-down analysis from yearly data to individual days.



Figure 3: Custom date hierarchy with Year, Quarter, Month and Day

5. Creating the Relationship Between Tables:

A relationship was established between the Dim_Date table and the Shopify stock table using their respective date columns.

The relationship was configured as:

- **Cardinality:** One-to-Many
- **Dim_Date:** One side
- **Shopify Stock table:** Many side
- **Cross-filter direction:** Single
- **Relationship:** Active

This relationship allows the date dimension to filter the stock data and provides a proper structure for time-based calculations.

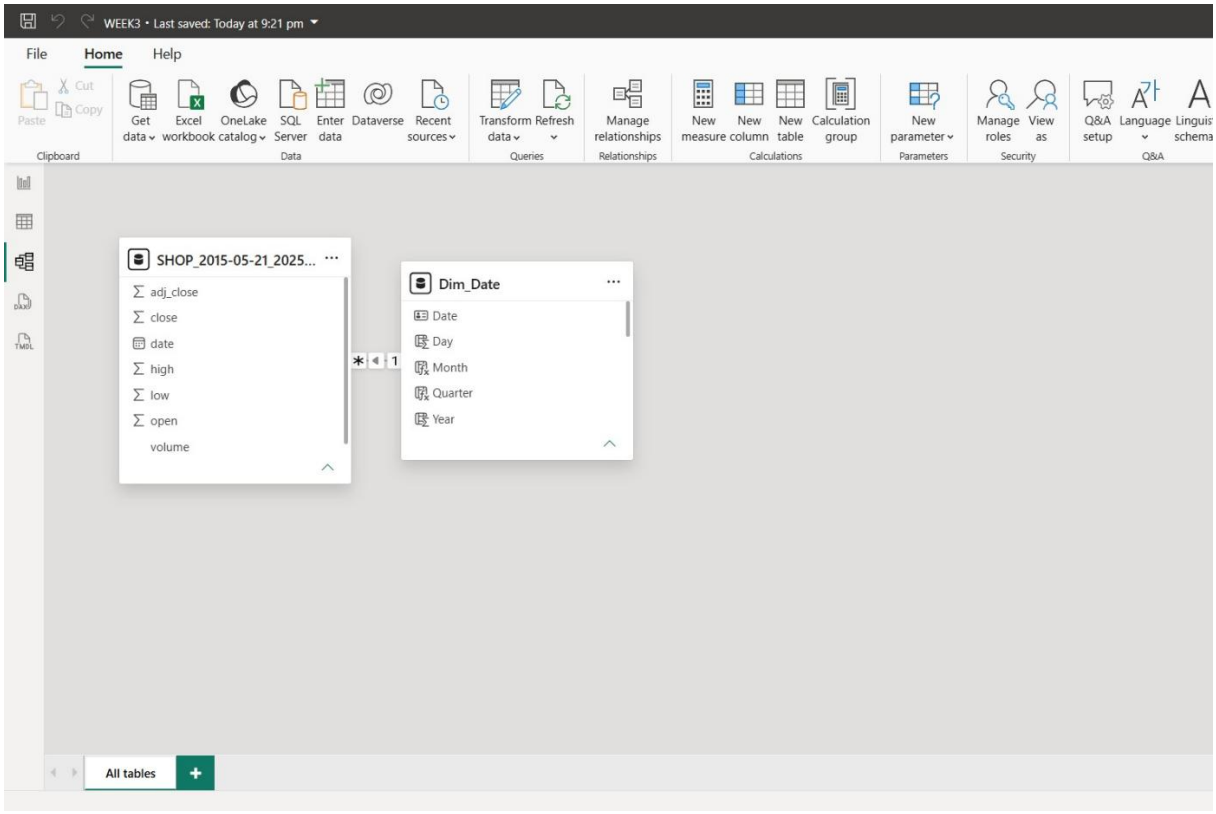


Figure 4: Active one-to-many relationship between Dim_Date and the Shopify Stock table

6. Creating Financial DAX Measures

Several DAX measures were created to perform financial calculations on the Shopify stock data.

The following core measures were created:

- **Open Price**
- **Close Price**
- **High Price**
- **Low Price**
- **Trading Volume**
- **Daily Return %**

The price measures were calculated using the average of their respective stock price columns.

Open Price

```
Open Price = AVERAGE('SHOP_2015-05-21_2025-03-16'[open])
```

Close Price

```
Close Price = AVERAGE('SHOP_2015-05-21_2025-03-16'[close])
```

High Price

```
High Price = AVERAGE('SHOP_2015-05-21_2025-03-16'[high])
```

Low Price

```
Low Price = AVERAGE('SHOP_2015-05-21_2025-03-16'[low])
```

Trading Volume

```
Trading Volume = SUM('SHOP_2015-05-21_2025-03-16'[volume])
```

Daily Return %

```
Daily Return % =
DIVIDE(
    [Close Price] - [Open Price],
    [Open Price],
    0
```

)

The Daily Return % measure calculates the percentage change between the average closing price and average opening price within the current filter context.

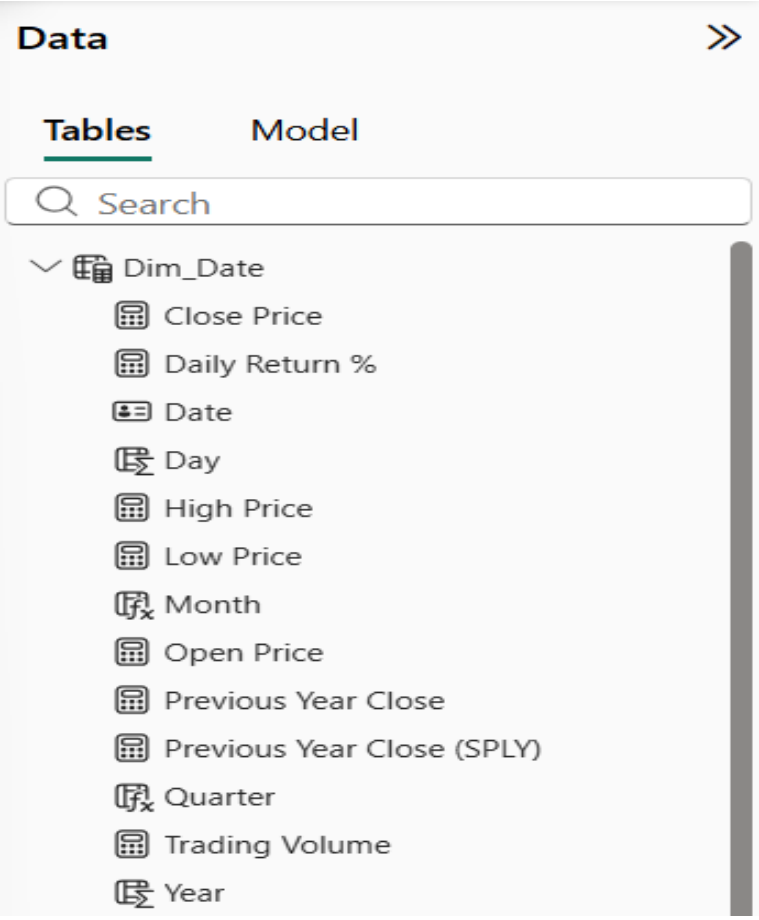


Figure 5: Core financial DAX measures created for Shopify stock analysis

7. Applying Time-Intelligence Functions:

Time-intelligence functions were implemented to perform calculations involving different time periods.

Three required functions were used:

- DATEADD

- TOTALYTD
- SAMEPERIODLASTYEAR

These calculations allow the stock data to be analysed across different time periods and compared with previous periods.

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a) DATEADD

A **Previous Year Close** measure was created using the DATEADD function.

```
Previous Year Close =
CALCULATE(
    [Close Price],
    DATEADD('Dim_Date'[Date], -1, YEAR)
)
```

This shifts the date context by one year and calculates the corresponding previous-year close price.

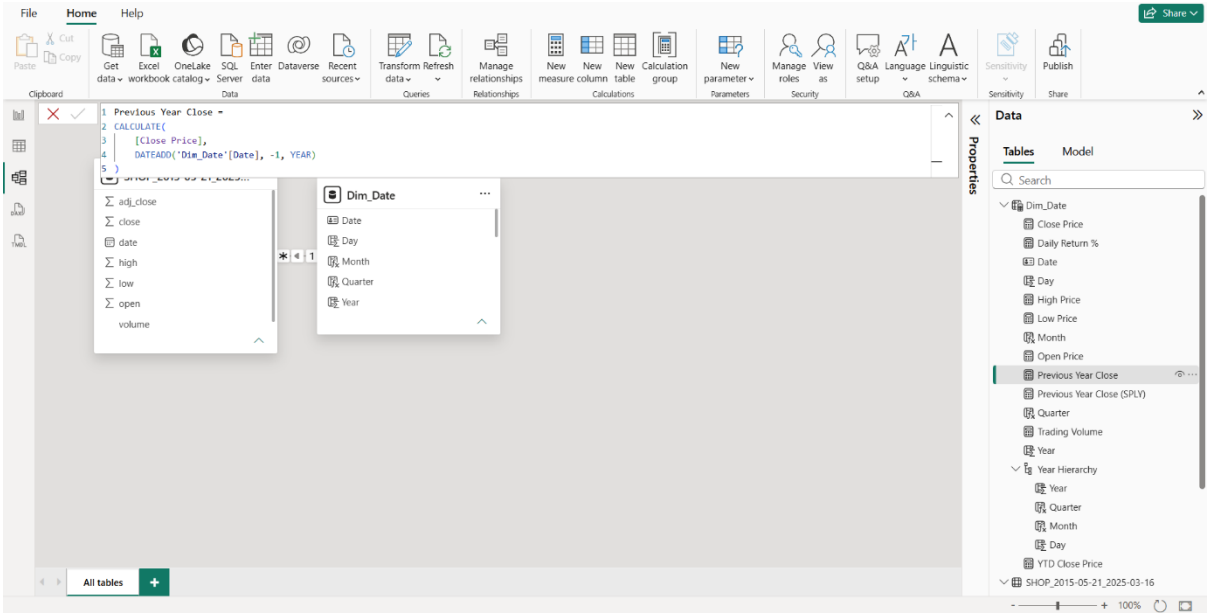


Figure 6: Previous Year Close measure using the DATEADD function

b) TOTALYTD

A **YTD Close Price** measure was created using the TOTALYTD function.

```
YTD Close Price =
TOTALYTD(
    [Close Price],
    'Dim_Date'[Date]
)
```

This calculates the year-to-date value of the Close Price measure based on the date context.

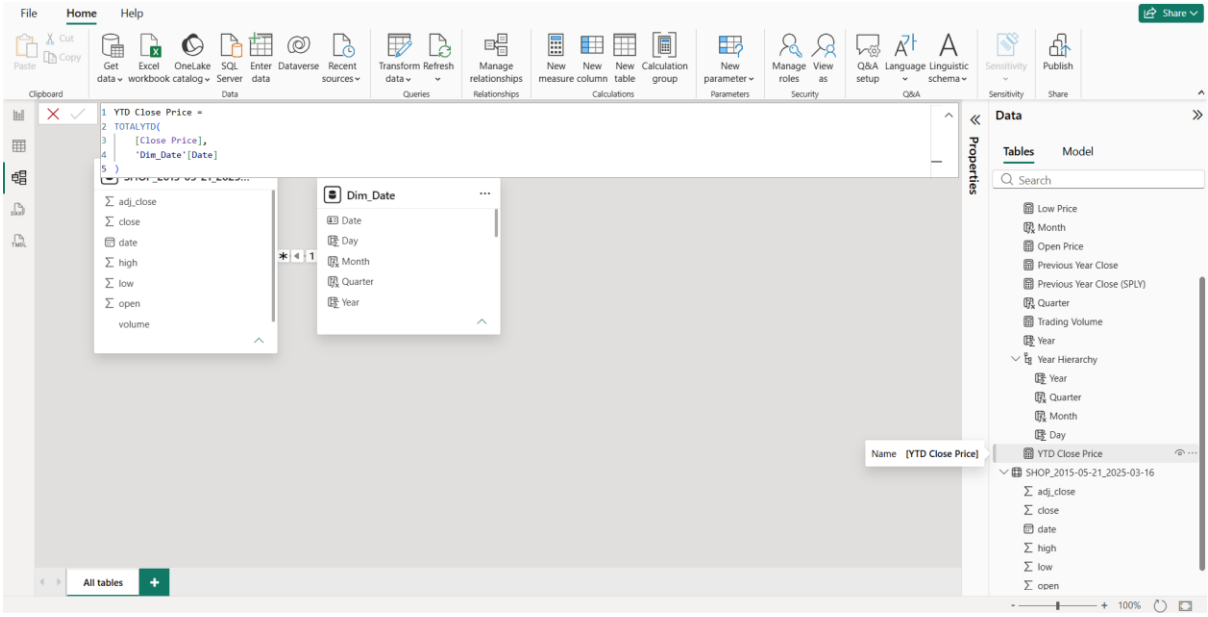


Figure 7: YTD Close Price measure using the TOTALYTD function

c) SAMEPERIODLASTYEAR

A **Previous Year Close (SPLY)** measure was created using the SAMEPERIODLASTYEAR function.

```
Previous Year Close (SPLY) =
CALCULATE(
    [Close Price],
    SAMEPERIODLASTYEAR('Dim_Date'[Date])
)
```

This allows the current period's Close Price to be compared with the corresponding period from the previous year.

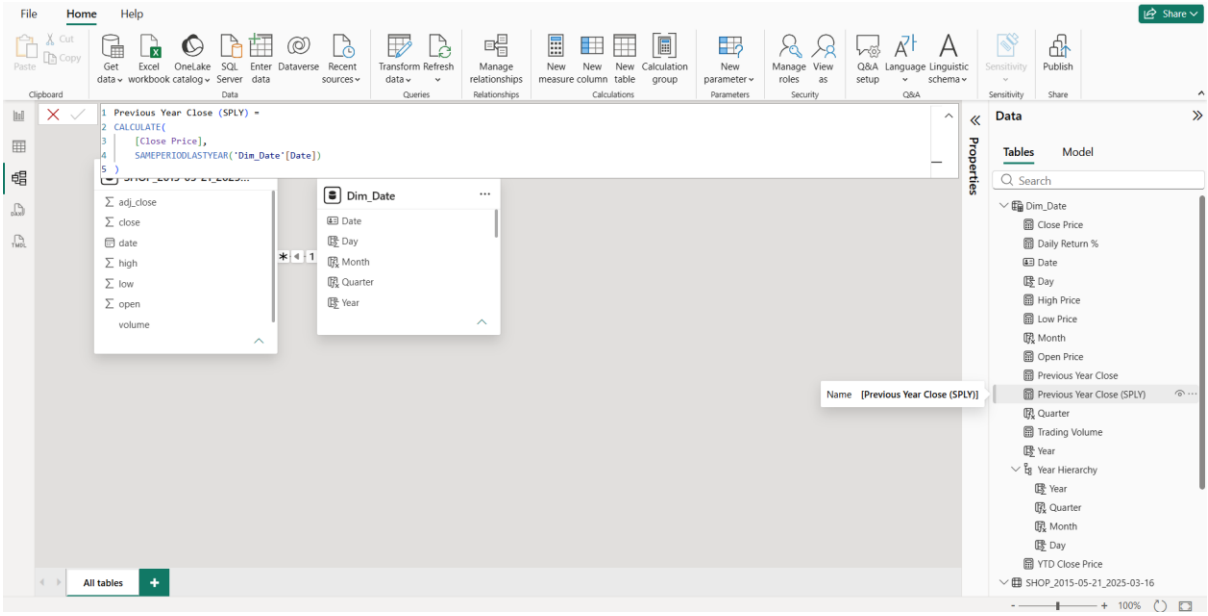


Figure 8: Previous Year Close measure using the SAMEPERIODLASTYEAR function

8. Final Data Model:

The completed Power BI model contains the Shopify Stock table and the dedicated Dim_Date table connected through an active one-to-many relationship.

The Dim_Date table contains the Date, Year, Quarter, Month and Day attributes along with the custom Year Hierarchy. The financial DAX measures and time-intelligence measures were also created within the model.

The final structure provides a foundation for financial analysis of Shopify stock prices, trading volume and changes across different time periods.

9. Conclusion:

Week 3 provided practical experience in preparing and modelling stock-market data using Power BI. The Shopify stock dataset was imported and prepared in Power Query, with the date column converted into the required Date format.

A dedicated Dim_Date table was created using CALENDARAUTO() and marked as the official Date Table. A one-to-many relationship was established between the date dimension and the Shopify Stock table, and a custom Year → Quarter → Month → Day hierarchy was created.

Financial DAX measures were developed for Open Price, Close Price, High Price, Low Price, Trading Volume and Daily Return %. Time-intelligence calculations using DATEADD, TOTALYTD and SAMEPERIODLASTYEAR were also implemented. Overall, the completed model provides a structured base for further financial analysis of Shopify stock data.